

N+ - Network Fundamentals and building network

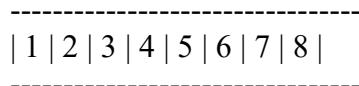
❖ Ethernet standard and specification

[Q-1] Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

[ANS]

Pin	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

➤ RJ-45 Connector (Front View)



[Q-2] Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

[ANS]

Standard	Speed	Cable Type	Maximum Length	Connector
10BASE-T	10 Mbps	Cat3/Cat5	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e/Cat6	100 m	RJ-45

**[Q-3] Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.**

[ANS]

- Cable Used:
 - Crossover Cable
- Reason:
 - A crossover cable swaps the transmit (TX) and receive (RX) wire pairs, allowing two similar devices (laptop-to-laptop) to communicate directly without a switch or hub.

[Q-4] List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

[ANS]

Device	Likely Ethernet Standard
WiFi Router	1000BASE-T
Desktop Computer (College Lab)	1000BASE-T
Network Printer	100BASE-TX